

Labor Market Facts: Stocks, Flows, and Measurement

Pedro Silos

Temple University

Fall 2026

Why Start with Measurement?

- Every model we build this quarter is disciplined by a small set of labor market facts.
- The facts are about **flows**, not just stocks.
 - The U.S. labor market reallocates workers at a staggering pace even when the unemployment rate barely moves.
- A stock can tell us *where* the economy is; only flows can tell us *why* it moves.

Goals of This Topic

1. Definitions and data sources: CPS, JOLTS, establishment data.
2. The lake model: unemployment as a flow steady state.
3. The “ins and outs” debate: what moves unemployment over the cycle?
4. Job flows vs. worker flows; vacancies and the Beveridge curve.

Readings

- *Macroeconomics* (free PhD textbook), Ch. 20, opening sections.
- Ljungqvist and Sargent, §6.4.
- Davis, Faberman, and Haltiwanger (2006).
- Shimer (2012).
- Elsby, Michaels, and Ratner (2015).

Labor Market States: CPS Definitions

Every civilian, non-institutional person aged 16 or older is in exactly one state each month:

- **Employed (E):** did any paid work in the reference week (including self-employment and temporary absence).
- **Unemployed (U):** no job, *actively searched* in the last four weeks, and available to work (plus those on temporary layoff awaiting recall).
- **Not in the labor force (N):** everyone else: students, retirees, home production, discouraged workers.

Source: BLS, Current Population Survey (CPS); monthly survey of roughly 60,000 households.

Derived Objects

$$u = \frac{U}{E + U}, \quad \text{LFPR} = \frac{E + U}{POP}, \quad \frac{E}{POP} = \text{employment--population ratio.}$$

- Note the judgment calls baked in: “actively searched” separates U from N .
- Model counterparts must respect this.
 - Search is an *activity*, not a label.

Headline Stock Magnitudes (U.S., Recent, Approximate)

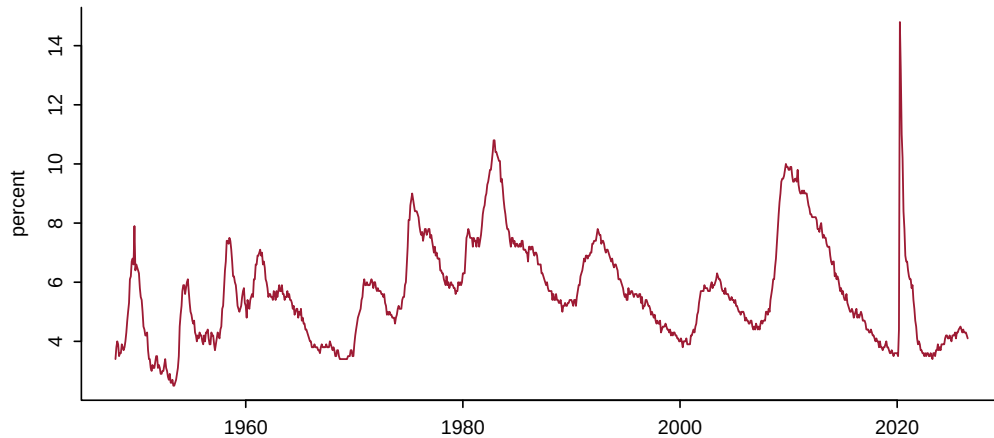
	Level	Rate
Civilian noninstitutional population, 16+	~270 m	—
Employment (E)	~160 m	$E/POP \approx 60\%$
Unemployment (U)	~7 m	$u \approx 4\text{--}4.5\%$
Labor force ($E + U$)	~168 m	LFPR $\approx 62.5\%$

Source: BLS, Current Population Survey, Table A-1. Update with the latest monthly release.

Unemployment and Participation over the Postwar Period

- Postwar average unemployment rate: roughly 5.7 percent.
 - Range from 2.5 percent (1953) to 10.8 percent (November 1982) and 14.7 percent (April 2020, a single month).
- Labor force participation rose from about 59 percent in the 1960s to about 67 percent in 2000 with women's entry.
 - Declining since, largely because of demographics (aging).

The U.S. Unemployment Rate, 1948–2025



Three Time-Series Facts about the Unemployment Rate

1. No trend, high persistence.
2. Asymmetry over the cycle.
3. Countercyclical and roughly coincident.

Fact 1: No Trend, High Persistence

- Unlike output, u_t is stationary.
- Monthly autocorrelation near 0.99.
- Business-cycle movements are large: the standard deviation of the cyclical component is roughly 10 times that of labor productivity (Shimer 2005).
 - A number to remember over the next couple of weeks.

Fact 2: Asymmetry

- Unemployment rises fast in recessions and declines slowly in expansions.
 - “Jobless recoveries” after 1991, 2001, and 2009.
- Plucking and rocket-and-feather dynamics are a challenge for linear models.

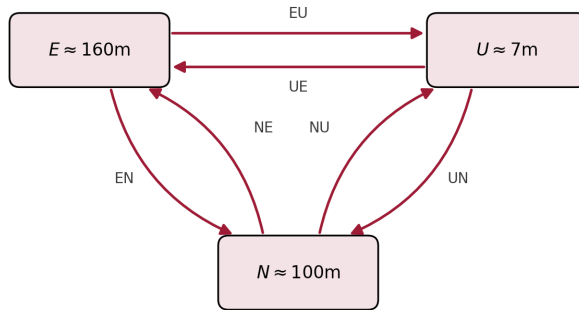
Fact 3: Countercyclical and Coincident

- u_t is strongly countercyclical and roughly coincident; vacancies lead slightly.
- The $u-v$ relationship (the Beveridge curve) is one of the tightest in macroeconomics.
- A stock, however, cannot tell us *why* it moves. For that we need flows.

The Iceberg Fact: Gross Flows Dwarf Net Changes

- In a typical month, U.S. employment changes on net by ± 0.1 – 0.2 million.
- Underneath that: roughly **5 million hires and 5 million separations per month** (JOLTS), and gross CPS flows across E , U , and N of similar magnitude.
- Metaphor: the unemployment pool is a lake with a nearly constant water level but enormous inflow and outflow.

Gross Worker Flows across E , U , and N



CPS Gross Flows: Monthly Transition Probabilities Average monthly probability of moving from row state to column state (approximate, U.S.):

From \ To	<i>E</i>	<i>U</i>	<i>N</i>
<i>E</i>	0.96	0.014	0.028
<i>U</i>	0.26	0.52	0.22
<i>N</i>	0.045	0.026	0.93

Source: approximate averages from published CPS gross flows research (BLS research series; Elsby, Hobijn, and Şahin 2015). Update with current data.

Reading the Transition Matrix

- **Unemployment is a short-duration state on average:** nearly half of the unemployed exit within a month (to E or N).
- **The $E \rightarrow U$ hazard is small but applied to a huge stock:** $0.014 \times 160\text{m} \approx 2.2\text{m}$ inflows per month.
 - Similar magnitude to $U \rightarrow E$ outflows, as steady state requires.
- **The participation margin is not negligible:** $U \rightarrow N$ and $N \rightarrow U$ flows are as large as flows involving E .
 - Two-state models abstract from this; keep it in mind.

Incidence vs. Duration: Two Views of the Same Pool

Two statements that sound contradictory are both true:

1. **Most unemployment spells are short.** Median completed duration in normal times is about 8–10 weeks; a large share end within a month.
2. **Most unemployment (person-weeks) is accounted for by long spells.** The few workers with long durations dominate the stock at a point in time (length-biased sampling).

Duration Dependence

- Long-term share (27+ weeks) of the unemployed: about 15–20 percent in normal times; peaked above 45 percent in 2010–11.
- Exit hazards decline with duration (*negative duration dependence*).
 - True dependence (skill decay, stigma, discouragement) or dynamic selection (heterogeneous types)?
 - Identification is hard; both matter.
- Models with a single exit hazard (the baseline lake and DMP models) miss this by construction.
 - Worth remembering when we calibrate.

See Machin–Manning-style duration decompositions; Kroft, Lange, and Notowidigdo (2013) on duration stigma

Setup: Two States, Two Hazards

- Fixed labor force of size 1. Two states: employed and unemployed. Time is discrete (monthly).
- **Separation hazard** s : probability an employed worker loses her job this month.
- **Job-finding hazard** f : probability an unemployed worker finds a job this month.
- For now (s, f) are exogenous constants.
 - The whole point of the DMP model is to make f , and later s , equilibrium objects.

Law of Motion for the Unemployment Rate

$$u_{t+1} = u_t + \underbrace{s(1 - u_t)}_{\text{inflows}} - \underbrace{fu_t}_{\text{outflows}} .$$

- Inflows: the employed who separate this month.
- Outflows: the unemployed who find a job this month.

Steady State

Setting $u_{t+1} = u_t$:

$$u^* = \frac{s}{s + f}.$$

- With $s = 0.02$ and $f = 0.40$ (rough U.S. monthly averages): $u^* = 0.02/0.42 \approx 4.8$ percent.
- Two small numbers, one familiar answer.

Dynamics

- Deviations decay geometrically:

$$u_{t+1} - u^* = (1 - s - f)(u_t - u^*).$$

- Convergence rate is $s + f \approx 0.42$ per month.
- Half-life:

$$\frac{\ln 2}{\ln\left(\frac{1}{1-s-f}\right)} \approx 1.3 \text{ months.}$$

The Punchline

- The U.S. labor market converges to its flow steady state within a quarter.
- To a first approximation, period by period,

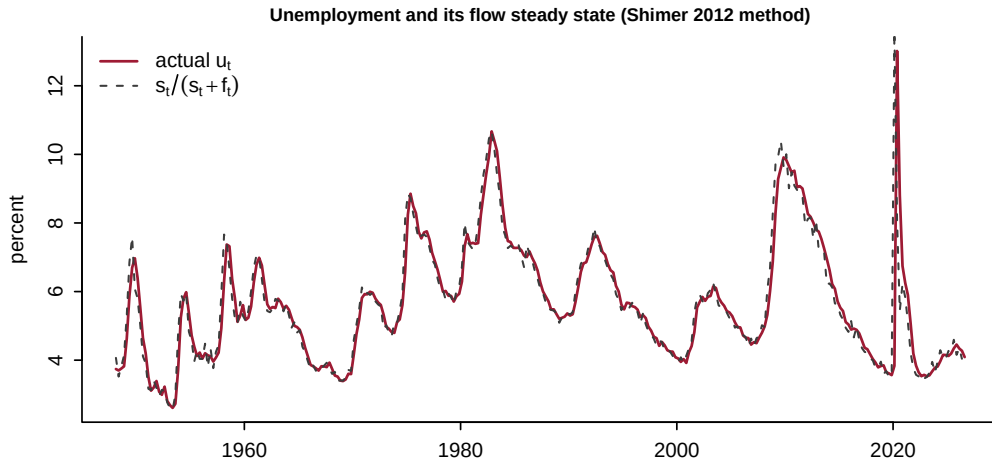
$$u_t \approx u_t^* = \frac{s_t}{s_t + f_t}.$$

- *Understanding unemployment means understanding the hazards s_t and f_t .*

Actual u_t vs. Flow Steady State $s_t/(s_t + f_t)$



The Measured Series



Three-State Lake Model

With states $\{E, U, N\}$ and monthly transition matrix P (rows sum to 1), the distribution $\pi_t = (\pi_t^E, \pi_t^U, \pi_t^N)$ evolves as

$$\pi_{t+1} = \pi_t P,$$

and the stationary distribution solves $\pi^* = \pi^* P$.

- π^* is the left eigenvector for eigenvalue 1.
- Existence and uniqueness follow from Markov chain theory (Ljungqvist and Sargent, Ch. 2).

Using the Three-State Model

- Feeding the CPS transition matrix into $\pi^* = \pi^* P$ reproduces the observed stocks ($u \approx 4\text{--}5$ percent, LFPR $\approx 62\text{--}63$ percent): a useful consistency check.
- The two-state model is the special case that consolidates N into the other states.
 - A good approximation for cyclical questions ($E \leftrightarrow U$ flows dominate cyclical variation);
 - a bad one for trend questions (participation).
- Growth in the labor force (entry b , exit d) is easily added; see the QuantEcon lecture.

Measuring the Hazards (Shimer 2012)

The CPS publishes stocks; Shimer's insight is that stocks at two dates identify the hazards.

- Let U_{t+1}^s be the short-term unemployed (duration less than 5 weeks) at $t + 1$: these are the period- t *inflows* still unemployed.
- Everyone in U_t who did not exit is in $U_{t+1} - U_{t+1}^s$, so

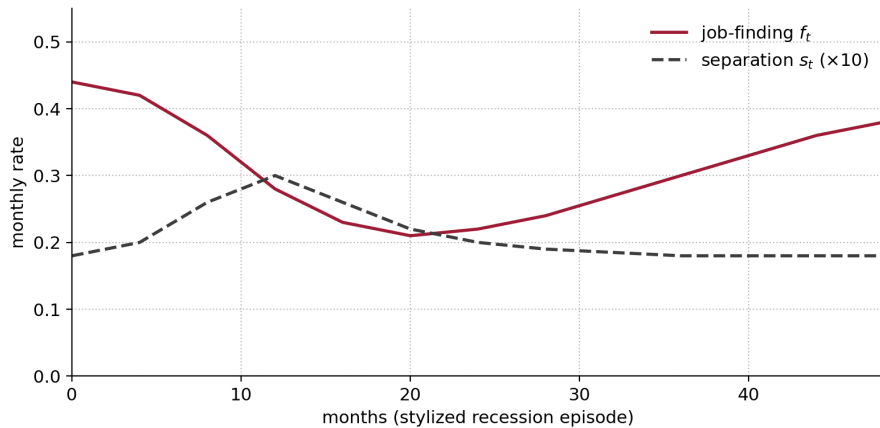
$$F_t = 1 - \frac{U_{t+1} - U_{t+1}^s}{U_t}, \quad f_t = -\ln(1 - F_t).$$

Separations and Time Aggregation

- **Separation.** Given f_t , back out s_t from the law of motion of u_t (or from short-term unemployment directly).
- **Time aggregation.** A worker can lose a job and find a new one within the month and never appear unemployed.
 - Discrete monthly probabilities understate high-frequency hazards;
 - Shimer's continuous-time correction handles this;
 - ignoring it overstates the role of separations.

Shimer (2012, *Review of Economic Dynamics*). Assignment 1 implements this on CPS data.

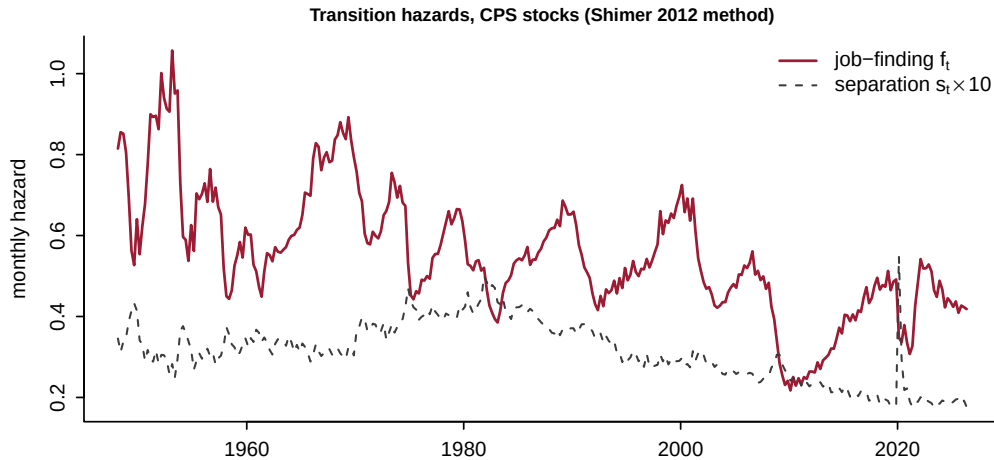
The Hazards over the Cycle



Reading the Hazards

- f_t swings from about 0.45 in booms to about 0.20 in deep slumps: large, persistent, procyclical.
- s_t (around 0.02) *spikes* at recession onsets, then reverts quickly.

The Measured Hazards



Who Moves Unemployment? The Decomposition Debate Using $u_t \approx s_t/(s_t + f_t)$, decompose $\text{var}(u)$ into contributions of f_t and s_t :

Study	Job-finding margin	Separation margin
Shimer (2012), U.S.	about 3/4	about 1/4
Fujita and Ramey (2009)	about 1/2	about 1/2
Elsby, Michaels, and Solon (2009)	about 2/3, but timing matters	spikes lead
Anglophone vs. continental Europe	finding dominates	larger role for inflows

Reading the Decomposition Debate

- Differences come from detrending, time aggregation, and steady-state vs. exact decompositions.
 - A live example of measurement choices driving conclusions.
- Working consensus: **the job-finding margin dominates at business-cycle frequencies**, but separations matter, especially in the first months of recessions.
- Consequence for theory: the baseline DMP model with *constant* s and endogenous f is a defensible first pass.
 - That is where Topic 2 starts.

Job Creation and Destruction (Davis–Haltiwanger)

Establishment-level data (BED, BDS, LBD) let us measure flows of *positions* rather than people.

For establishment e with employment n_{et} :

$$JC_t = \sum_{e: \Delta n_{et} > 0} \Delta n_{et}, \quad JD_t = \sum_{e: \Delta n_{et} < 0} |\Delta n_{et}|.$$

- **Job reallocation** = $JC + JD$.
- **Net growth** = $JC - JD$.
- **Excess reallocation** = $JC + JD - |JC - JD|$: churn beyond what net change requires.

Job Flow Magnitudes

- Approximate U.S. magnitudes: JC and JD are each about 6–8 percent of employment *per quarter*.
 - Roughly one in four jobs is created or destroyed within a year.
- Most reallocation is *within* narrowly defined industries and regions: idiosyncratic, not sectoral.
 - Heterogeneity is the rule, a theme that returns in Topics 4–5.

Worker Flows = Job Flows + Churn	
Flow (per month, U.S., approximate)	Size
Hires (JOLTS)	~5.5m (~3.7% of E)
Total separations (JOLTS)	~5.3m
of which: quits	~3m (procyclical)
of which: layoffs and discharges	~1.7m (countercyclical)
Job creation + destruction (monthly equivalent)	~4m

Source: BLS, Job Openings and Labor Turnover Survey (JOLTS); BED. Update with the latest release.

Replacement Hiring and Quits

- Hires exceed job creation: firms replace quitters even at constant employment (*replacement hiring*).
 - Worker reallocation is roughly twice job reallocation.
- Quits are procyclical (job-ladder climbing, the on-the-job search extensions of Topic 2); layoffs are countercyclical.
- The *quit rate* is one of the best real-time indicators of labor market tightness.

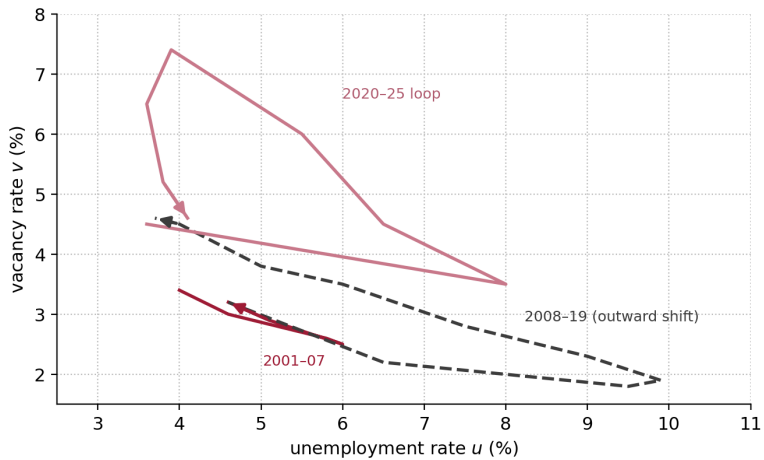
Vacancies

- A **vacancy** (JOLTS): a specific position exists, could start within 30 days, and the employer is actively recruiting.
 - Measured monthly since December 2000; before that, proxied by the Conference Board help-wanted index.
- Vacancy rate $v = V/(E + V)$: ranged from roughly 2–3 percent in slumps to 7 percent at the 2022 peak.
- Strongly *procyclical*: the mirror image of unemployment.

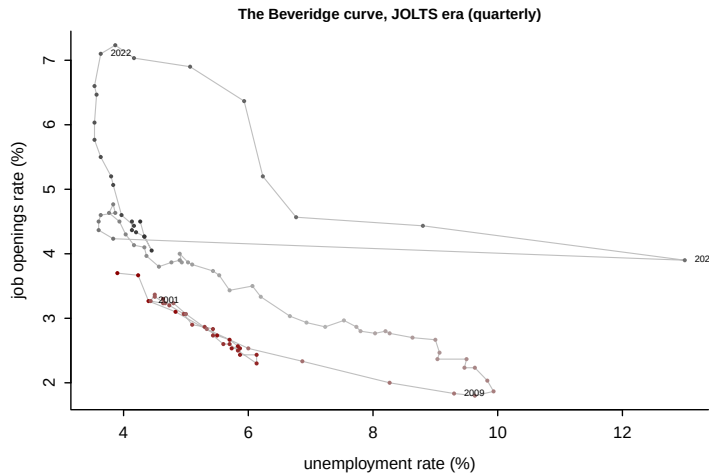
Labor Market Tightness

- The ratio $\theta = v/u$, **labor market tightness**, will be the single state variable of the DMP model.
- θ moved from about 0.15 (2009) to about 2 (2022): an order of magnitude.
- Whatever theory we write must produce large equilibrium swings in θ from plausible shocks.

The Beveridge Curve



The Measured Beveridge Curve



Reading the Beveridge Curve: Slope and Loops

- **Why downward sloping?** Recessions bring fewer postings and longer queues: high u , low v .
 - Any theory with a stable matching technology and cyclical demand for hires delivers this.
- **Why loops?** u is a stock that adjusts slowly (half-life of about 1–3 months) while v is a jump variable.
 - Counterclockwise dynamics around turning points.

Reading the Beveridge Curve: Shifts

- 2009–13 outward shift: extended UI? mismatch (construction)? duration composition?
- Post-2021 inward re-normalization from a very tight market.
- The decomposition matters for policy: a shift means higher unemployment *at given tightness*, that is, lower matching efficiency.
- The Beveridge curve is the empirical face of the **matching function**, to which we now turn.

The Aggregate Matching Function

Posit that meetings are produced from searchers and vacancies:

$$M_t = m(U_t, V_t) = A U_t^\alpha V_t^{1-\alpha} \quad (\text{CRS, Cobb-Douglas}),$$

so the hazards inherit tightness $\theta = V/U$:

$$f(\theta) = \frac{M}{U} = A \theta^{1-\alpha}, \quad q(\theta) = \frac{M}{V} = A \theta^{-\alpha}, \quad f(\theta) = \theta q(\theta).$$

Interpreting the Matching Function

- Estimates: elasticity of matches with respect to unemployment $\alpha \approx 0.5\text{--}0.7$ (Petrongolo and Pissarides 2001 survey).
 - CRS is hard to reject at the aggregate level.
- Interpretation: a *black box* standing in for coordination frictions, information, heterogeneity, geography.
 - Micro-foundations exist (urn-ball, stock-flow, directed search) and imply different cyclical behavior.
- This one function converts the flow accounting of this topic into an *equilibrium theory* once V is chosen by firms. That is the DMP model.

Heterogeneity in Unemployment (Levels, Not Cycles) Unemployment rates differ persistently across groups (U.S., late 2010s, approximate):

Education	<i>u</i>	Age	<i>u</i>
Less than high school	~5.5%	16–24	~8–9%
High school	~3.8%	25–54	~3%
Some college	~3.0%	55+	~2.5%
Bachelor's degree or more	~2.1%		

Source: approximate; update from CPS Tables A-1 and A-4. See Cajner et al. (2017) and Elsby, Hobijn, and Şahin (2010) on group flow decompositions.

Reading the Group Differences

- Group gaps are driven mostly by *inflow rates* (separations), not by job-finding rates.
 - Low-education and young workers lose jobs far more often.
- Cyclical sensitivity also differs: the same aggregate shock roughly *scales* group unemployment rates (Okun's "high-pressure economy" arithmetic).
- These facts motivate the heterogeneous-agent machinery of Topics 4–5: who bears unemployment risk matters for precautionary savings and welfare.

Temporary page!

$\text{\LaTeX}$ was unable to guess the total number of pages correctly. As there was some unprocessed data that should have been added to the final page this extra page has been added to receive it. If you rerun the document (without altering it) this surplus page will go away, because $\text{\LaTeX}$ now knows how many pages to expect for this document.